

[Share Your Experience](#): Take the 2024 Developer Survey

Get 95% confidence interval with glm(.) in R

Asked 10 years, 5 months ago Modified 10 years, 5 months ago Viewed 54k times Part of R Language Collective



Here are some data

18

```
dat = data.frame(y = c(9,7,7,7,5,6,4,6,3,5,1,5), x = c(1,1,2,2,3,3,4,4,5,5,6,6), color = rep(c('a','b'),6))
```



and the plot of these data if you wish



```
require(ggplot)
ggplot(dat, aes(x=x,y=y, color=color)) + geom_point() + geom_smooth(method='lm')
```

When running a model with the function `MCMCglmm()` ...

```
require(MCMCglmm)
summary(MCMCglmm(fixed = y~x/color, data=dat))
```

I get the lower and upper 95% interval for the estimate allowing me to know if the two slopes (color = a and color = b) are significantly different.

When looking at this output...

```
summary(glm(y~x/color, data=dat))
```

... I can't see the confidence interval!

My question is:

How can I have these lower and upper 95% interval confidence for the estimates when using the function `glm()` ?[R](#) [r](#) [statistics](#) [glm](#) [confidence-interval](#) [mixed-models](#) [Edit tags](#)[Share](#) [Edit](#) [Follow](#) [Close](#) [Flag](#)

asked Dec 16, 2013 at 20:29



Remi.b

17.9k ● 32 ● 91 ● 171

There is a `confint` function in the **MASS** package, I believe. – joran Dec 16, 2013 at 20:33

...also, be careful judging statistical significance by comparing confidence intervals. You might miss some significant results. – joran Dec 16, 2013 at 20:36



In this case, they work: just look at whether the confidence bounds for the "x:colorb" coefficient overlap 0. – alex keil Dec 16, 2013 at 20:41

1 "confint" is also in the stats package. – alex keil Dec 16, 2013 at 20:43

2 @alexkeil Yes, but the glm method is in **MASS**, as noted in `?confint`. – joran Dec 16, 2013 at 20:44

2 Answers

Sorted by:
[Reset to default](#)

Date modified (newest first)



USE `confint`

24

```
mod = glm(y~x/color, data=dat)
summary(mod)
Call:
glm(formula = y ~ x/color, data = dat)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.11722  -0.40952  -0.04908   0.32674   1.35531

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   8.8667     0.4782  18.540 0.0000000177
x            -1.2220     0.1341  -9.113 0.0000077075
x:colorb      0.4725     0.1077   4.387   0.00175

(Dispersion parameter for gaussian family taken to be 0.5277981)

Null deviance: 48.9167  on 11  degrees of freedom
Residual deviance: 4.7502  on 9  degrees of freedom
AIC: 30.934

Number of Fisher Scoring iterations: 2

confint(mod)
Waiting for profiling to be done...
      2.5 %      97.5 %
(Intercept)  7.9293355  9.8039978
x            -1.4847882 -0.9591679
x:colorb      0.2614333  0.6836217
```

Share Edit Follow Flag

edited Dec 16, 2013 at 21:48

answered Dec 16, 2013 at 20:34

alex keil
1,021 ● 7 ● 14

▲ Why is no one asking whether there is justification for not including colorb as a main effect? – IRTFM Dec 16, 2013 at 21:12

▲ @DWin I'd like to compare how the levels of `color` interact with the effect of `x` on `y`. I am only interested in comparing slopes. I don't really care about the overall effect of `color`. Does it sound like a good justification? – Remi.b Dec 16, 2013 at 21:31

1 ▲ Not to me. Unless you offer evidence that color alone has no effect, then you are *reducing* your power to study an interaction by constraining the color main effect to be zero. – IRTFM Dec 16, 2013 at 21:34

▲ @Remi.b, look up "Interaction contrast ratio" for a better way to address the statistical significance (or, even better, estimation) of the degree to which there is interaction in your data (in certain models). – alex keil Dec 16, 2013 at 21:39

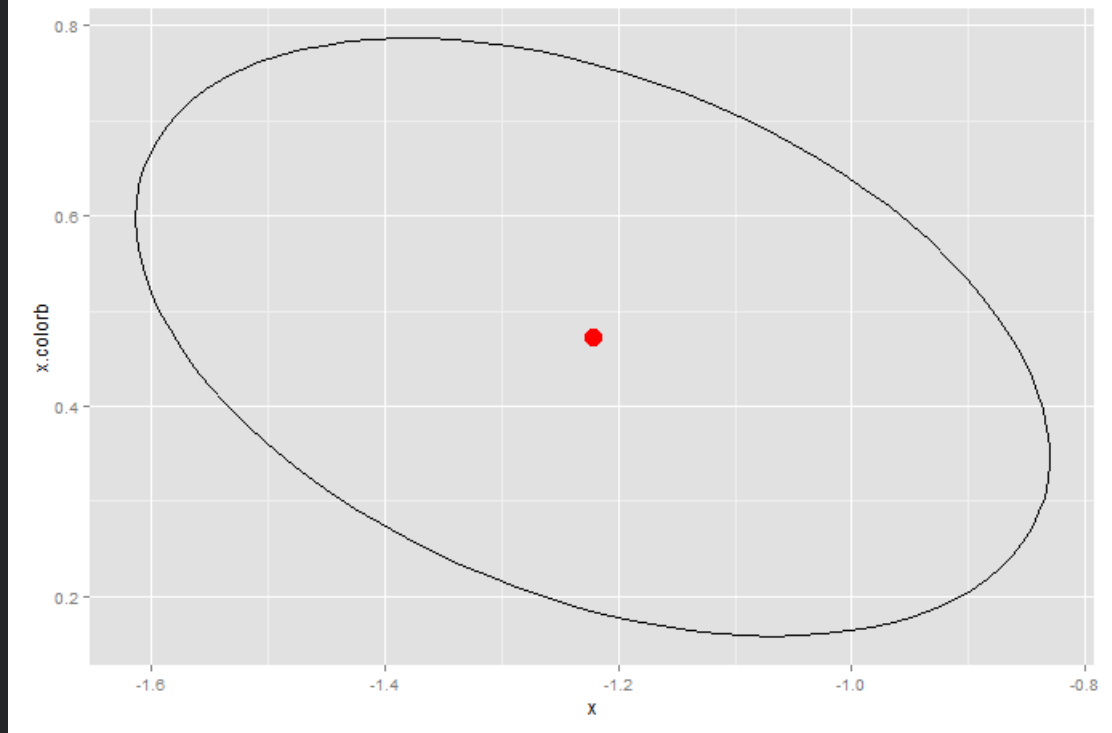
1 ▲ @Drubio $1 - 0.95 = 0.05$, which corresponds to 5% of the distribution. Because you want a two tailed confidence limit you divide the .05 in half and look at where it cuts but bottom 2.5% and top 2.5% of the distribution. If you look at a picture of the distribution, you would scan from left to right you cut off the lowest 2.5% of the distribution and when you get to 97.5% you cut again and take everything to the right. – itsMelnMiami Jul 20, 2020 at 11:17

▲ @alex's approach will get you the confidence limits, but be careful about interpretation. Since glm is fundamentally a non-linear model, the coefficients usually have large covariance. You should at least take a look at the 95% confidence ellipse.

6

```
mod <- glm(y~x/color, data=dat)
require(ellipse)
conf.ellipse <- data.frame(ellipse(mod, which=c(2,3)))
ggplot(conf.ellipse, aes(x=x, y=x:colorb)) +
  geom_path() +
  geom_point(x=mod$coefficient[2], y=mod$coefficient[3], size=5, color="red")
```

Produces this, which is the 95% confidence ellipse for x and the interaction term.



Notice how the confidence limits produced by `confint(...)` are well with the ellipse. In that sense, the ellipse provides a more conservative estimate of the confidence limits.

Share Edit Follow Flag

answered Dec 16, 2013 at 21:11



jlhoward

58.9k ● 7 ● 100 ● 142

- ▲ There are a lot of ways to come up with more conservative confidence bounds. What are the methods used in the ggplot package?
 ▼ That's unclear. Profile bounds have really nice properties and are not subject to the shortcoming that you noted. They are often used in non-linear models. Since the original model used GLM with ID link and gaussian distribution, there is no need to worry about the "fundamental non-linearity." – alex keil Dec 16, 2013 at 21:33
- ▲ `ggplot` is just used for plotting. `ellipse(...)` comes from the ellipse package. From the plot, it is clear that `x` and `x:colorb` are strongly correlated. – jlhoward Dec 16, 2013 at 21:36
- ▲ The coefficients are correlated, but I don't understand why that is an argument against testing whether a single coefficient is 0.
 ▼ – alex keil Dec 16, 2013 at 21:43
- 1 ▲ In this case, neither the horizontal ("x") or vertical ("x:colorb") axis crosses the ellipse, so you can assert that both coefficients are different from 0 at $p=0.05$. If one or the other axis *did* cross the ellipse, but the limits from `confint(...)` did not cross 0, then I would be reluctant to make that assertion. As I understand it, `confint(...)` produces limits based on the *assumption* that covariance is 0. – jlhoward Dec 16, 2013 at 21:51
- ▲ That's a joint test of two coefficients - how does that better answer the OP's question than a single coefficient? – alex keil Dec 16, 2013 at 22:03